

Cluster separation is quantified by the silhouette score (Rousseeuw, 1987) (range -1 to $+1$, higher value means tighter and better-separated clusters), computed in the native 4096-dimensional space; 2-D PCA is used for visualisation only.

4 Analysis and Discussion

4.1 Is Harmful-Refusal Mediated by a Single Universal Direction?

We replicate Arditi et al. (2024) to establish the baseline. This approach extracts a DIM direction from the residual stream and steers it to suppress refusal. For steering, their approach selects the layer at which the refusal direction is most linearly separable from harmless activations. We select the layer that maximises the projection score gap: for each candidate layer ℓ and direction $\hat{r}_\ell$, the projection score of a sample i is $s_i = \mathbf{h}_i^\ell \cdot \hat{r}_\ell$, and the gap is $\bar{s}_{\text{refused}} - \bar{s}_{\text{benign}}$, which is the difference in mean scores between refused-harmful and benign-answered populations. A larger gap indicates that the direction better discriminates between the two groups at that layer.

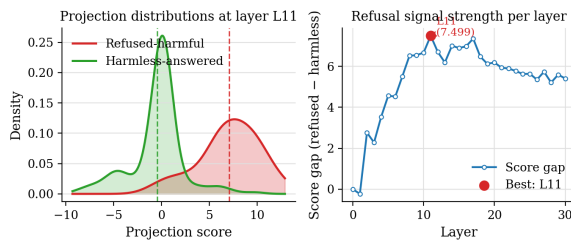


Figure 2: Harmful-refusal DIM direction selection. Left: projection score distributions at L11, where refused-harmful and benign-answered are cleanly separated. Right: score gap across layers, peaking at L11 and L17.

The score gap peaks at layers 11 and 17 (Figure 2, right), where the two groups separate cleanly with near-zero overlap (Figure 2, left). We extract the universal refusal direction $\hat{r}$ at layer 11 and apply global ablation steering (Table 2).

	Baseline	After Steering
Harmful-refusal rate	68%	20%
Benign refusal rate	0%	0%
Attack success rate	32%	80%

Table 2: Global refusal direction ablation. Here, we steer away from the universal refusal direction.

Steering the refusal direction in reverse, $\hat{r}$ does reduce the harmful-refusal rate from 68% to 20%,

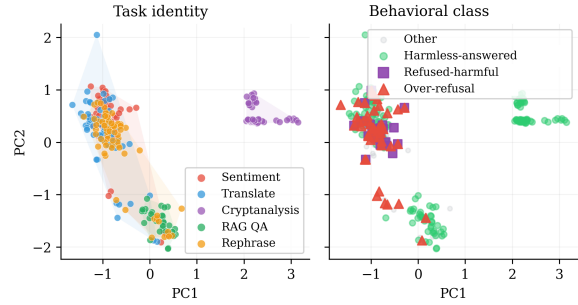


Figure 3: PCA at the peak task-identity layer (layer 12). Left: five task clusters. Right: over-refusal (triangles) sits overlapping within the non-refusal clusters.

while leaving the benign refusal rate unchanged. This works because harmful-refusal directions are highly consistent (and supposedly universal) (Section §4.2). A single vector suffices to steer responses from non-refusal to refusal; and harmful-refusal direction is task-agnostic.

4.2 Is Over-Refusal Mediated by a Single Direction?

Here, we demonstrate that over-refusal is task-dependent perturbation embedded within task-identity structure. Figure 3 analyses Layer 12 (where the task representation separability peaks) and shows that five task form clusters in the left panel, and in the right panel, the over-refusal samples (triangles) reside within their respective non-refusal task clusters rather than in a separate refusal space. To quantify how tightly over-refusal clusters within each task, we measure the centroid distance between over-refusal and benign-answered samples for each task and compare it to the global pooled refusal setting (Figure 4). Per-task centroid distances are consistently lower than the global setting, suggesting that over-refusal is a within-cluster perturbation rather than a cross-task signal. The global refusal distance is much higher because pooling across tasks mixes distinct geometries, inflating apparent separation.

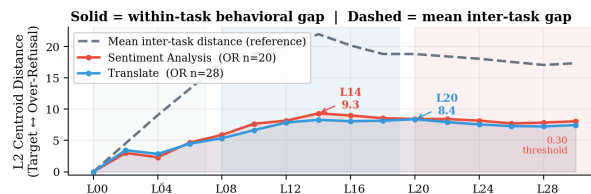


Figure 4: Per-task centroid distances vs. inter-task / global centroid distances across layers. The gap suggests that over-refusal is a within-cluster perturbation and is less likely to be task-agnostic.

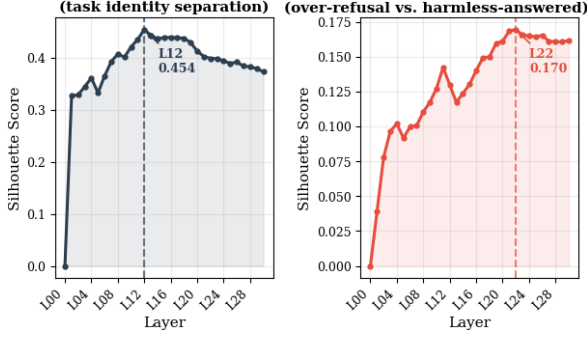


Figure 5: Inter-task silhouette (left) peaks at layer 12; per-task behavioural silhouette (right) peaks later and much weaker, suggesting that task structures form in the early-mid layers.

We further measure cluster quality using silhouette scores, where we compare each sample’s distance to its own cluster against its distance to the nearest other cluster, higher scores indicating better separation (Figure 5). Inter-task silhouette peaks strongly at layer 12 (0.45), while per-task silhouette peaks later but remains substantially weaker (0.17). This suggests task structure crystallises earlier and more sharply than any refusal-type separation. We note similarities with the findings of (Lindsey et al., 2025), which identify task-specific SAE features emerging in intermediate layers, while refusal-related features appear predominantly in later layers.

No shared over-refusal direction. To test whether over-refusal has a consistent direction with harmful-refusal, we compute the cosine similarity between the global harmful-refusal DIM direction ($\hat{\mathbf{r}}_\ell$ from §4.1) and the mean over-refusal direction ($\hat{\mathbf{v}}_\ell^{\text{OR}}$) layer by layer (Figure 6). The over-refusal direction plateaus with the harmful-refusal, with only partial overlap (cosine similarity of 0.47), enough to cause incidental refusal suppression under global ablation but not enough for precise steering. We then inspect the per-task over-refusal directions in embedding space (Figure 7) and note that universal and task-specific directions slightly align at mid-layers, but the task-wise directions are noticeably shorter, different for each task, and remain consistent within the same task.

Over-refusal spans a higher-dimensional subspace. We analyse the residual-stream activations of each refusal samples at each layer. Explained variance estimates how much of the total activation variance is captured by a given number of principal components (Figure 8). A population whose vari-

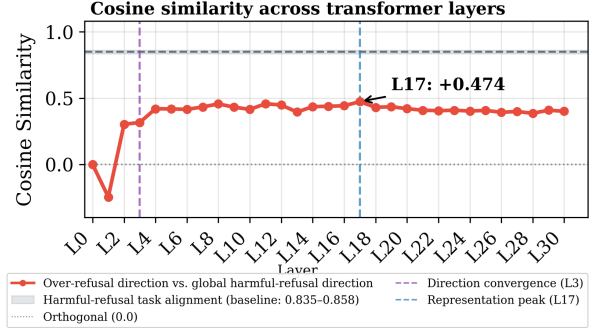


Figure 6: Layer-wise cosine between OR direction ($\hat{\mathbf{v}}_\ell^{\text{OR}}$) and universal refusal direction ($\hat{\mathbf{r}}_\ell$). The over-refusal direction plateaus well below the harmful-refusal reference (0.85).

ance is concentrated in few components has a compact, low-dimensional subspace amenable to single-direction interventions. We track n_{80} , the number of components needed to reach 80% explained variance. Although harmful-refusal concentrates more variance in its leading component, over-refusal requires substantially more components to reach the same threshold, and this gap persists throughout mid-layers. This suggests that smaller set of directions may not sufficiently capture the over-refusal signal, and any low-rank ablation targets the wrong axes.

4.3 Are Harmful-Refusal and Over-Refusal Linearly Separable?

We apply linear probing to analyse whether we can separate the representation of the refusal types. If the two refusal types occupy geometrically distinct regions, a linear classifier trained on residual-stream activations should be able to separate them. We train a logistic-regression probe at each layer on held-out samples, independently for three targets: (a) task identity (5 tasks); (b) over-refusal behaviour (over-refusal vs. benign-answered); and (c) refusal type (over-refusal vs. harmful-refusal).

Task identity saturates from the first layer (Figure 9a), which is expected as the five tasks share a common instruction template (e.g. “Translate the following sentence ...” for Translation task), so task-discriminative signal is already present in the input token embeddings before the substantial computation in the later layers.

Refusal behaviour consolidates more gradually, peaking in the mid-layers (Figure 9b). This is consistent with prior discussion on over-refusal being a within-task perturbation and that the model must first organise representations around task identity

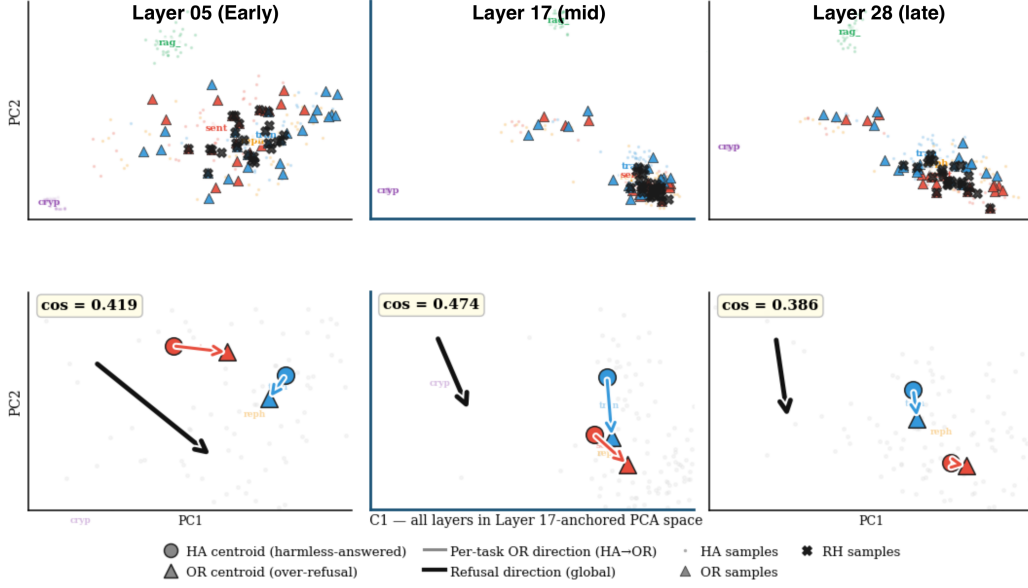


Figure 7: 2-D PCA at three representative layers. Top row: residual-stream activations of harmless / benign-answered (HA) and over-refusal (OR) samples with HA and OR centroids marked. Bottom row: the over-refusal DIM direction (HA→OR centroid) and the [Arditi et al. \(2024\)](#) harmful-refusal direction overlaid in the same plane (black arrow). The over-refusal direction diverges from the harmful-refusal direction and varies across layers.

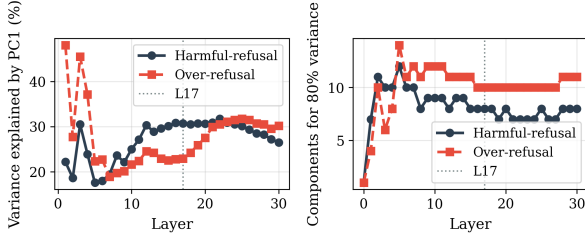


Figure 8: No. of Principal Components needed to reach 80% explained variance) across layers for over-refusal (red) and harmful-refusal (black). Over-refusal consistently requires more dimensions to reach the 80% threshold, although harmful-refusal has higher variance.

before the finer distinction between a compliant and an erroneously refused response becomes geometrically resolvable. The refusal-type probe achieves seemingly high accuracy from layer 1 (Figure 9c). Harmful instructions are lexically distinct from sensitively benign task inputs, and the input texts come from different dataset distributions, hence the early separability considers surface-level input differences, rather than a learned representational distinction. So, probing alone may not sufficiently distinguish these refusals, and we further explore using causal patching in Section 4.6.

4.4 Does Global Ablation Suppress Over-Refusal?

We apply universal refusal steering $\hat{\mathbf{r}}_\ell$ to the over-refusal samples and measure the refusal rate (Fig-

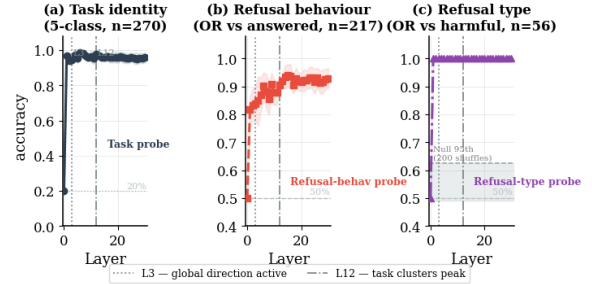


Figure 9: Linear probe accuracy across layers. (a) Task identity saturates from early layers. (b) Over-refusal behaviour (over-refusal vs. benign-answered) consolidates through mid-layers. (c) Refusal type (over-refusal vs. harmful-refusal) perfect accuracy from early layers.

ure 10). Global ablation suppresses both refusal types, but suppresses harmful refusals more than over-refusal. The suppression ratio (over-refusal reduction divided by harmful-refusal reduction) falls below 1 (Table 3): the intervention disrupts the safety refusals more than it addresses erroneous refusals. The partial overlap between $\hat{\mathbf{r}}_\ell$ and $\hat{\mathbf{v}}_\ell^{\text{OR}}$ causes incidental over-refusal suppression; the lack of full alignment suggests that the ablation is not well-suited for over-refusal mitigation.

4.5 Does Task-Conditioned Steering Outperform Global Ablation?

Following our hypothesis, a task-conditioned steering method that operates within task-dependent